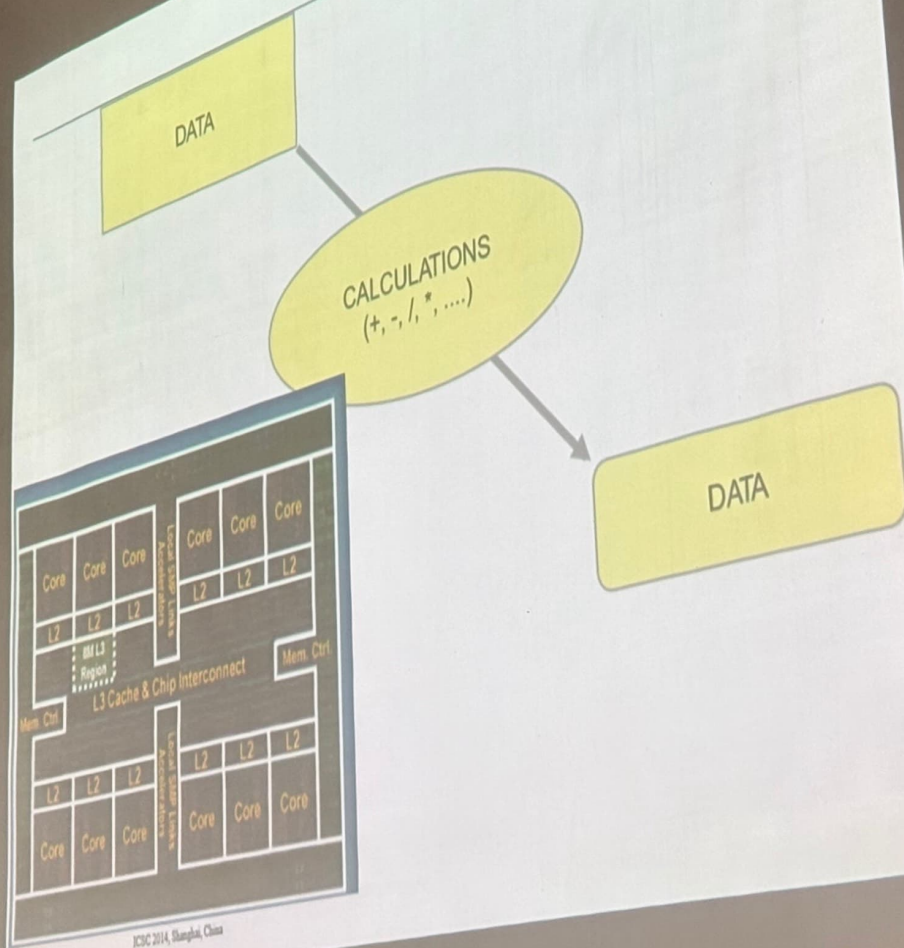


The Roofline Model



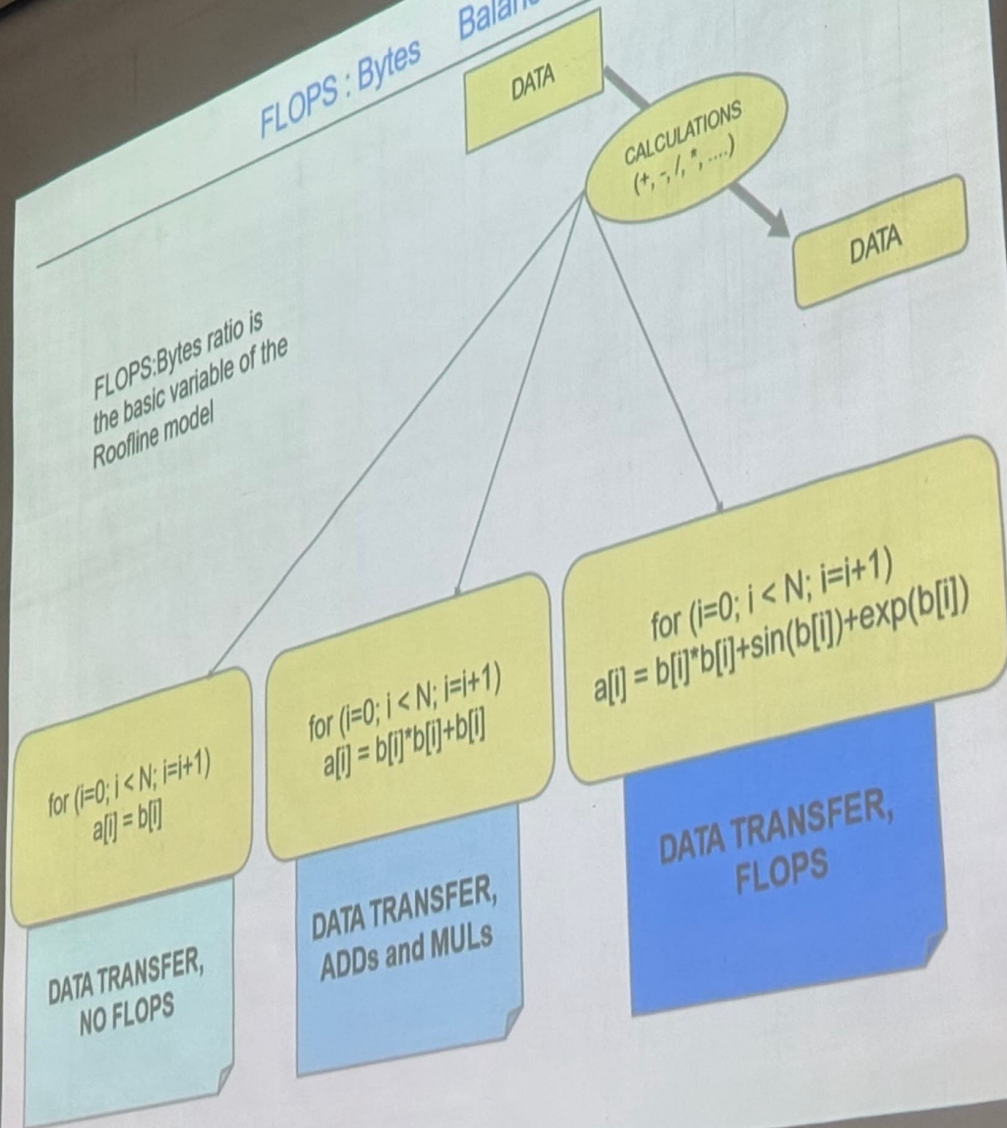
The Roofline Model

- The roofline model was introduced in 2009 by Williams et.al.
 - Samuel Williams, Andrew Waterman, and David Patterson. 2009. Roofline: an insightful visual performance model for multicore architectures. Commun. ACM 52, 4 (April 2009), 65-76. DOI=10.1145/1498765.1498785 <http://doi.acm.org/10.1145/1498765.1498785>
- It provides an easy way to get performance bounds for compute and memory bandwidth bound computations.
- It relies on the concept of Computational Intensity (CI) – sometimes also called Arithmetic or Operational Intensity.
- The Roofline Model provides a relatively simple way for performance estimates based on the computational kernel and hardware characteristics.

Performance [GF/s] = function (hardware and software characteristics)

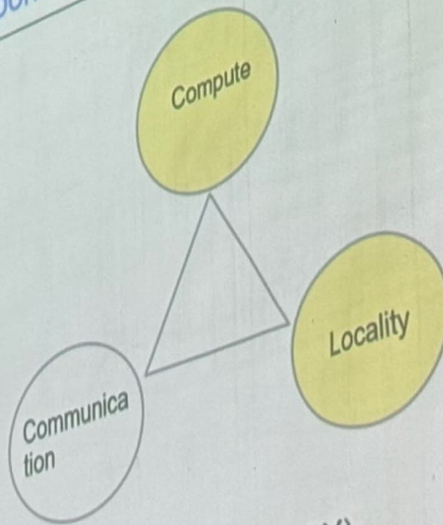
FLOPS : Bytes Balance

FLOPS:Bytes ratio is the basic variable of the Roofline model



The Roofline Model: Principal Components to Performance

Performance can be estimated
from hardware and kernel characteristics



Kernels can be Compute bounded (DGEMM) or Communication bounded (DAXPY)
(kernels are rarely well balanced)

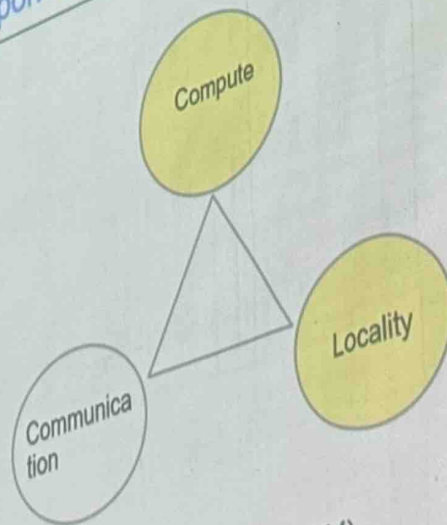
Some hardware is more communication oriented than another (high memory BW)

Some hardware is more computation oriented than another (high FLOPs)

Mapping kernel characteristics to hardware characteristics (or vice-versa) → performance

The Roofline Model: Principal Components to Performance

Performance can be estimated
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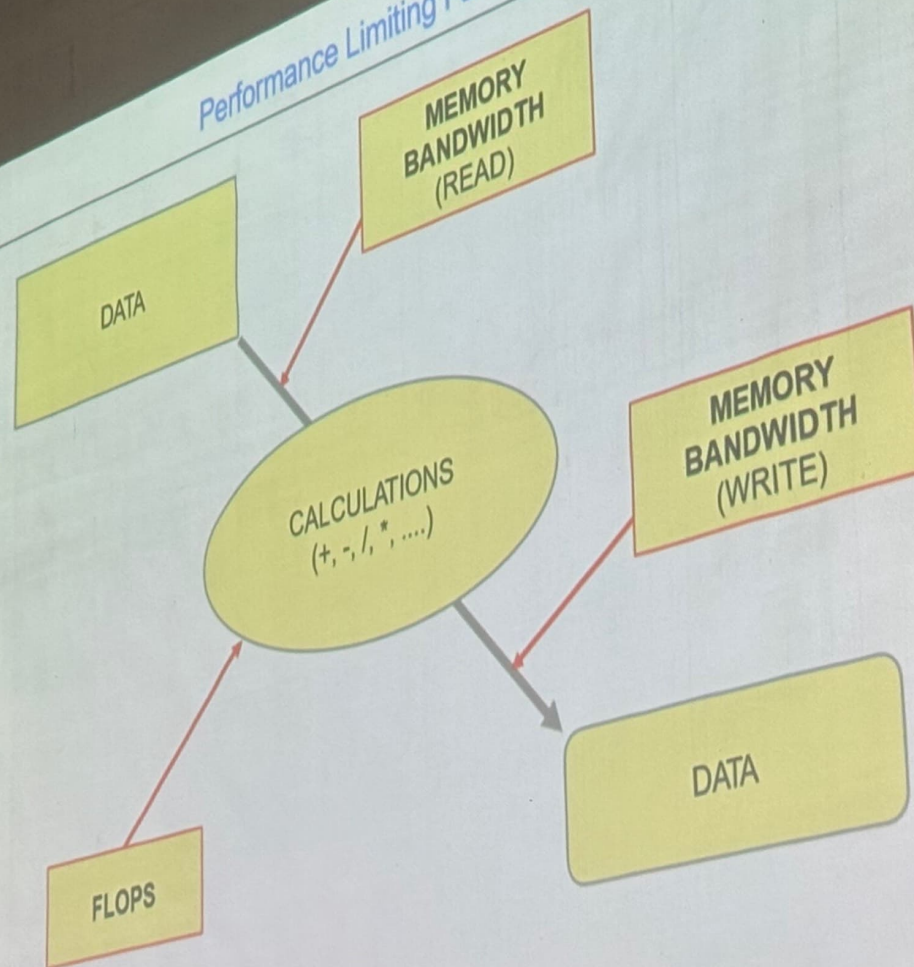
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Performance Limiting Factors

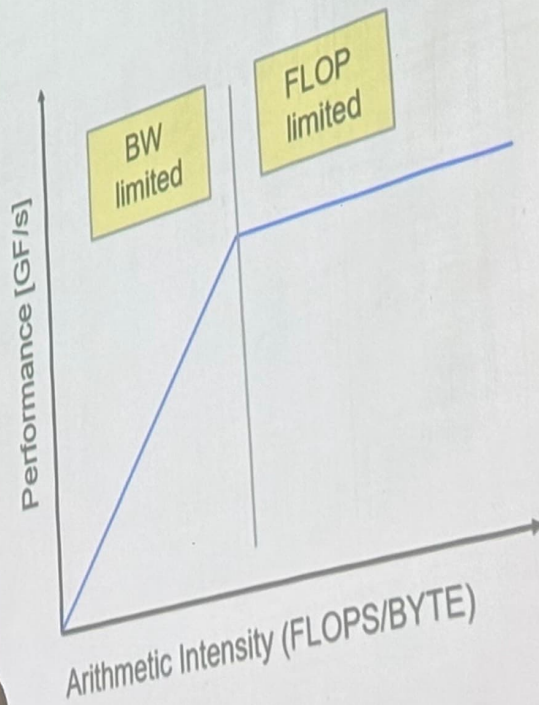


The Roofline Model

The Roofline Model - is a tool to understand the kernel/hardware limitation and it is also a tool for kernel optimization

Performance is upper bounded by:

- 1) the peak flop rate
- 2) the streaming bandwidth

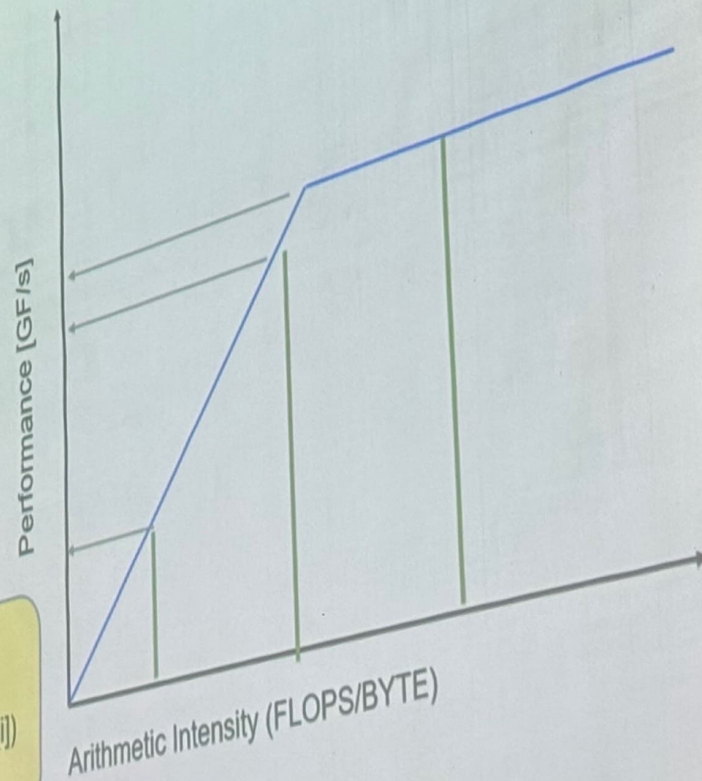


The Roofline Model

for ($i=0$; $i < N$; $i=i+1$)
 $a[i] = 2.3 * b[i]$

for ($i=0$; $i < N$; $i=i+1$)
 $a[i] = b[i] * b[i] + b[i]$

for ($i=0$; $i < N$; $i=i+1$)
 $a[i] = b[i] * b[i] + \sin(b[i]) + \exp(b[i])$



The Roofline Model: Arithmetic Intensity (AI)

FLOPS / Bytes ratio – one of the basic characteristics of a kernel

```
for (i = 0; i < N; ++i)
  z[i] = x[i] + y[i]
```

2 (8 byte) loads
1 (8 byte) write
ADD
 $AI = 1 / (2 \cdot 8 + 8) = 1/24^*$

```
for (i = 0; i < N; ++i)
  z[i] = x[i] + y[i] * x[i]
```

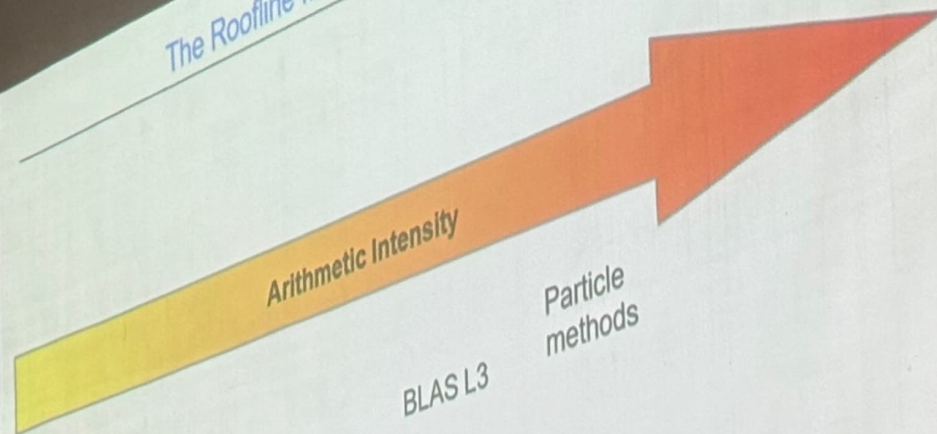
2 (8 byte) loads
1 (8 byte) write
ADD
MUL
 $AI = 2 / (2 \cdot 8 + 8) = 1/12^*$

```
for (i = 0; i < N; ++i){
  l1 = A_offset[i]; l2 = A_offset[i+1];
  sum = 0.0
  for (j = 0; j < (l2-l1); ++j)
    sum += A[l1+j] * x[col_index[l2+j]];
  y[i] = sum;
}
```

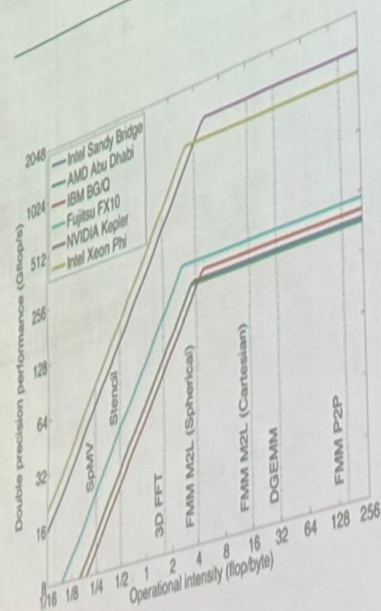
2 (8 byte) + 1 (4 bytes) loads
1 (8 byte) write
ADD
MUL
 $AI = 2 / (2 \cdot 8 + 4 + 8) = 1/14$

* because of write-allocate traffic on cache-based systems kernel would actually requires an extra read for Z and have even lower AI.

The Roofline Model: Arithmetic Intensity (AI)



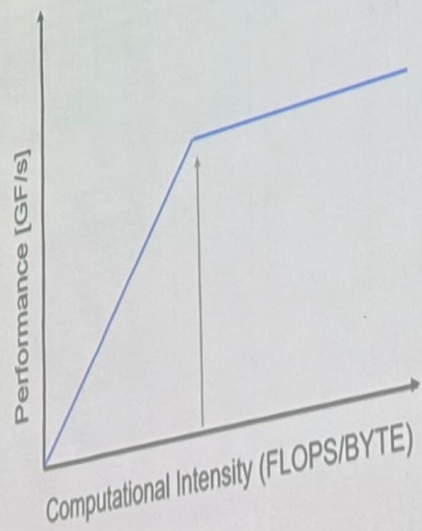
The Roofline Model: Kernel-Hardware mapping



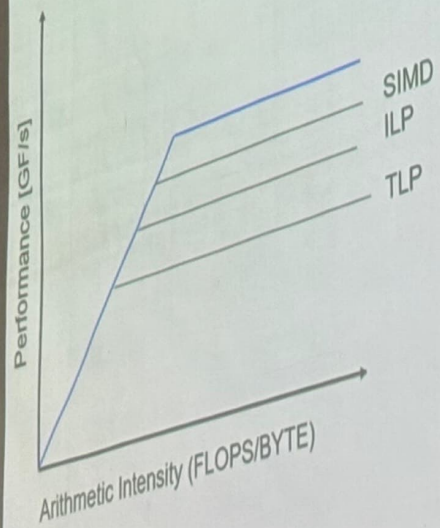
- ▣ The trend is for architectures to have ever decreasing machine balance (the point where the bandwidth roof meets the ceiling moves to the right).
- ▣ More and more algorithms are going to find themselves memory bound.
- ▣ Even DGEMM can run into trouble depending on the blocking factor chosen.
- ▣ A "balanced" architecture can also be a "crippled" one, e.g. low-end GPUs with 1/24th the DP peak performance.
- You can achieve a higher percentage of a lower peak.

How Will the Fast Multipole Method Fare in the Exascale Era?
 SIAM News, Volume 46, Number 6, July/August 2013
 By Lorena A. Barba and Rio Yokota (Boston University & KAUST)

It is an art to find a perfect match between kernel and hardware characteristics
In another words it requires a lot of work to create a kernel that will exhaust both, the memory BW and FLOPs capacity at the same time. (many times it is even impossible)



The Roofline Model: Performance Limiting Factors

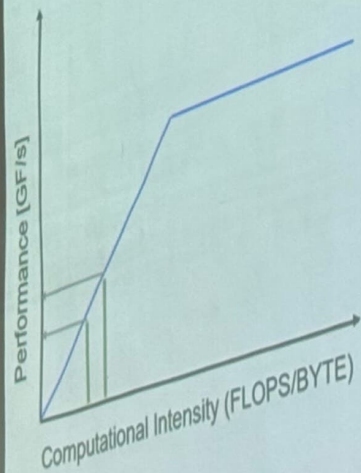


Performance depends on how well a given kernel fits node/processor architecture,

and/or how well a given kernel is translated by a compiler.

Recall: hardware-kernel characteristics mapping.

The Roofline Model: Performance limiting factors



N is large, i.e., buffer does not fit cache

```
for (i=0; i < N; ++i)
    a[i] = buffer[i] + b[i];
```

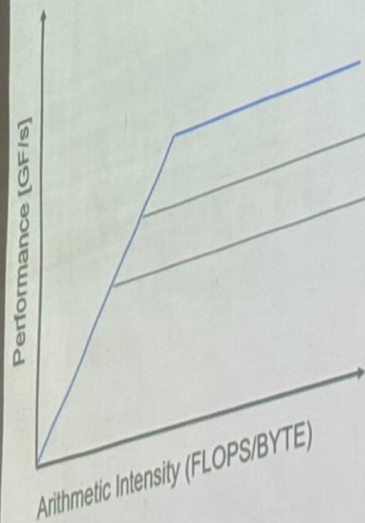
```
for (i=0; i < N; ++i)
    c[i] = buffer[i] + d[i];
```

$$AI_{total} = 2 / (2 * 3 * 8) = 1/24;$$

```
for (i=0; i < N; ++i){
    a[i] = buffer[i] + b[i];
    c[i] = buffer[i] + d[i];
}
```

$$AI = 2 / (5 * 8) = 1 / 20;$$

The Roofline Model: Performance Limiting Factors - Instruction Level Parallelism (ILP)



```
sum = 0;
for (i=0; i < N; ++i)
    sum = sum + a[i];
```

```
sum0 = sum1 = sum2 = sum3 = 0;
for (i=0; i < N; i+=4){
    sum0 = sum0 + a[i];
    sum1 = sum1 + a[i+1];
    sum2 = sum2 + a[i+2];
    sum3 = sum3 + a[i+3];
}
sum0 = sum0+sum1;
sum2 = sum2+sum3;
sum = sum0+sum2;
```

Example 1: DAXPY

Consider DAXPY : for ($i = 0; i < N; ++i$) $y[i] = a * x[i] + y[i]$

For each "i" : 1 addition, 1 multiplication
2 loads of 8 bytes each
1 store

Execution on BlueGene/Q (Peak 204.8 GFLOP/node)

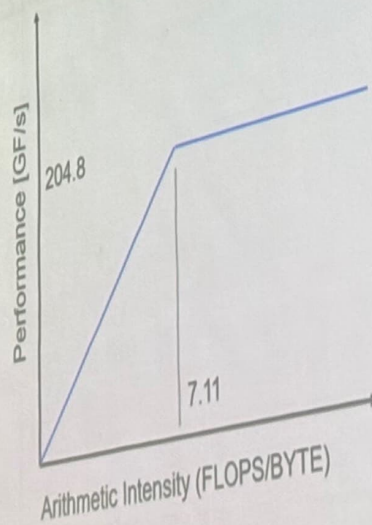
Performance estimates:

$$AI = 2 / (3 * 8) = 1 / 12$$

$1/12 < 7.11 \rightarrow$
We are in the memory BW
limited area on the
Roofline plot

$$7.11 / (1 / 12) = 85.32$$

$$204.8 / 85.32 = 2.4 \text{ GF/s}$$



Example 1: DAXPY

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Execution on BlueGene/Q (Peak 204.8 GFLOP/node):

# threads	Time [s]	GFLOPS	DDR traffic per node (Bytes/cycle)
1	0.0879111	0.455	3.519
2	0.044039	0.907	7.022
4	0.022151	1.801	13.94
8	0.0174019	2.284	17.686
16	0.017447	2.287	17.719

Performance estimates:

$$AI = 2 / (3 * 8) = 1 / 12$$

$$1/12 < 7 \rightarrow$$

We are in the memory BW
limited area on the roofline
plot

$$7.11 / (1 / 12) = 85.32$$

$$204.8 / 85.32 = 2.4 \text{ GF/s}$$